Ansh Goyal

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Summary

Third-year ECE undergraduate specializing in Industrial AI, Edge ML, and AI engineering, with hands-on research deployed on live ONGC Solar Turbine equipment. Builds production-grade agentic AI (RAG pipelines, multi-agent orchestration, MCP integration), edge computer-vision systems, and SHAP-interpretable forecasting models. Targeting IEEE Transactions on Industrial Informatics (SCI Q1).

Education

Jaypee Institute of Information Technology, Noida 2024 – 2028
B.Tech in Electronics & Communication Engineering CGPA: 8.31 — Sem 4 SG: 9.2
Coursework: Machine Learning, Deep Learning, Bayesian Inference, Probability & Statistics, Optimization Theory, Linear Algebra, DSA, Signal Processing, OOP.

Technical Skills

- **AI Engineering:** RAG Pipelines, LangChain, LlamaIndex, FAISS, ChromaDB, Pinecone, Vector Databases, MCP (Model Context Protocol), Agentic AI, Multi-Agent Systems, Tool Use, Function Calling, Embeddings
- **ML & AI:** PyTorch, Scikit-learn, XGBoost, LightGBM, Isolation Forest, Temporal Transformer, Conformal Prediction, SHAP, LSTM, CNN, Anomaly Detection, Time-Series Forecasting
- **Computer Vision:** YOLOv8, OpenCV, MediaPipe, NCNN (edge deployment), Roboflow
- **Languages & Tools:** Python, C++, SQL, Git, Streamlit, Jupyter, MATLAB, Groq API, LLaMA 3.3 70B, Raspberry Pi

Experience

Oil and Natural Gas Corporation (ONGC) - Research Intern, Predictive Maintenance May 2026 - Jul 2026
Solar Turbine Condition Monitoring

- Deployed a multi-scale feature-fusion pipeline on live ONGC Solar Turbine data (4-channel vibration, 42,698 points, 10-sec sampling): fused 6 detectors (Isolation Forest, Temporal Transformer, statistical baselines) via a conformal-prediction layer, raising an alert 57.5 hrs before ESD (peak vibration 75.0 mm/s, 223% of baseline; all 4 channels in alarm).
- Added SHAP root-cause analysis (DE-Y 35%, DE-X 27%, NDE-X 20%, NDE-Y 18%) with NL maintenance actions; validated on the NASA IMS benchmark (58-hr lead time, zero false alarms); targeting IEEE TII (SCI Q1).

Infosys Springboard (Virtual) — Machine Learning Intern Nov 2025 – Jan 2026

- Built an end-to-end CNN audio-classification pipeline with MFCC feature extraction for multi-class musical-instrument recognition.

Research & Publications

A Latch-On-Resistant, False-Alarm-Gated Lead-Time Metric for Bearing Anomaly Detection — and What It Reveals About SCADA-Rate Logging Submitted, 2026
Sole author. *International Journal of Prognostics and Health Management (IJPHM).* Introduces a false-alarm-gated lead-time metric for rolling-bearing prognostics; benchmarks 9 detectors across 4 run-to-failure datasets (XJTU-SY, FEMTO/PRONOSTIA, University of Ferrara, NASA IMS) plus a real ONGC gas-turbine record; full reproducible pipeline released on GitHub with an archived Zenodo DOI.

Projects

Research Agent *Python, LLaMA 3.3 70B, Groq, RAG, FAISS, LangChain, XGBoost, SHAP*
11-agent autonomous pipeline with RAG-powered retrieval (FAISS + arXiv/Semantic Scholar): topic → 20+ papers ingested → gap identification → ML experiments (RF, XGBoost, LightGBM; 5-fold CV, p-values, 95% CI) → complete IEEE-quality paper with 6 auto-generated figures; runs on Groq free tier at ∼20k tokens/run with 3 human approval checkpoints.

Real-Time Metal Crack Detection on Edge *YOLOv8, NCNN, Raspberry Pi 5*
... Trained YOLOv8 to 97.5% precision and 98% mAP@50; NCNN-exported with threaded inference at 30–40 FPS on a Raspberry Pi 5 under hard edge-hardware constraints.

Dynamic Pricing Engine for Urban Parking *Python, Pandas, NumPy, Bokeh, Pathway*
Data-driven pricing engine with 3 models (baseline linear, demand-based, competition-aware) on real urban parking-lot data; simulated real-time price updates rendered as interactive Bokeh dashboards tracking occupancy-to-price dynamics.

Explainable AI Traffic Management System *YOLOv8, LSTM, SHAP, Scikit-learn*
YOLOv8 vehicle detection → LSTM traffic forecasting → congestion classification (Low/Medium/High) with a SHAP interpretability layer; manuscript in preparation for an IEEE conference.

Leadership & Teaching

IEEE Student Branch, JIIT Noida — Organising Secretary June 2026 – Present
Lead technical event strategy and execution for a 500+ member IEEE branch; manage a cross-functional committee of 20+ members across workshops, seminars, and industry collaborations.

IEEE Student Branch, JIIT Noida — Workshop Mentor, Agentic AI Bootcamp Sept 2025
Designed and delivered an LLM & multi-agent architecture curriculum to 150+ students over 5 days; mentored project teams from concept to working prototype.